

5.2 Entropy (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.2 Entropy (A Level Only)
Difficulty	Hard

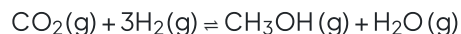
Time allowed: 50

Score: /38

Percentage: /100

Question 1a

Carbon dioxide and hydrogen can react to form methanol.



Data for enthalpy of formation and entropy is shown in Table 1.1.

Table 1.1

	CO₂(g)	H₂(g)	CH₃OH(g)	H₂O(g)
$\Delta H_f / \text{kJ mol}^{-1}$	-394	0	-201	-242
$S / \text{J K}^{-1} \text{mol}^{-1}$	214	131	238	189

Calculate the Gibbs free-energy change (ΔG), in kJ mol^{-1} , for this reaction at 790 K.

[4 marks]**Question 1b**

Explain whether this reaction is feasible at 790 K.

[1 mark]**Question 1c**

Use your values of ΔH and ΔS from part (a) to calculate the temperature above which this reaction is feasible.

[3 marks]

Question 1d

Explain why the reaction in part (a) has a negative entropy change.

[1 mark]

Question 2a

This question is about the feasibility of a reaction.

Calcium carbonate can be thermally decomposed to make calcium oxide.



Table 2.1 shows the standard entropies of each substance.

Table 2.1

Substance	$\text{CaCO}_3(\text{s})$	$\text{CaO}(\text{s})$	$\text{CO}_2(\text{g})$
$S / \text{JK}^{-1}\text{mol}^{-1}$	89	40	214

i)
Use the information in Table 2.1 to show that calcium carbonate is stable at room temperature (25 °C).

[4]

ii)
Calculate the minimum temperature needed to decompose calcium carbonate.

[2]

[6 marks]

Question 2b

Explain how the conditions can be changed for the reaction in (a) to make the reaction feasible.

[3 marks]**Question 2c**

The evaporation of water is an endothermic reaction but also a spontaneous reaction.

Explain why.

Refer to the entropy change that occurs and the change in the arrangement of water molecules.

[4 marks]

Question 2d

Use the information in Table 2.2 to calculate the temperature above which the reaction between hydrogen and oxygen to form gaseous water is **not** feasible.

Table 2.2

	$S^\ominus / \text{J K}^{-1} \text{mol}^{-1}$	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
$\text{H}_2(\text{g})$	131	0
$\text{O}_2(\text{g})$	205	0
$\text{H}_2\text{O}(\text{g})$	189	-242

[4 marks]**Question 2e**

Explain what would happen to the sample of gaseous water if it was heated to a higher temperature than that calculated in part (d).

[2 marks]

Question 3a

Free energy changes can be used to predict the feasibility of chemical reactions and processes.

Table 3.1 shows the equations of five different chemical reactions or processes.

Complete the table by predicting the sign of ΔS for each chemical reaction and process

Table 3.1

chemical reaction or process	sign of ΔS
$\text{MgCl}_2(\text{s}) + (\text{aq}) \rightarrow \text{MgCl}_2(\text{aq})$	
$\text{Ca}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{aq}) + \text{H}_2(\text{g})$	
$\text{Na}_2\text{SO}_4(\text{s}) + 10\text{H}_2\text{O}(\text{l}) \rightarrow \text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}(\text{s})$	
$2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g})$	
$\text{C}_2\text{H}_5\text{OH}(\text{g}) \rightarrow \text{C}_2\text{H}_5\text{OH}(\text{l})$	

[2 marks]**Question 3b**

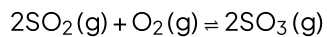
The enthalpy and entropy changes of a reaction are both negative.

Explain how the feasibility of this reaction will change as the temperature decreases.

[2 marks]

Question 3c

The equation a reversible reaction is shown below:



Use the information in Table 3.2 to determine the temperature at which the reaction becomes feasible. Show your working.

Table 3.2

	SO₂	O₂	SO₃
$\Delta H_f^\ominus / \text{kJ mol}^{-1}$	-296.8	0	-395.7
$S^\ominus / \text{JK}^{-1} \text{mol}^{-1}$	248.2	205.1	256.8

[6 marks]

Model Answers: Hard

1a

a) To calculate the Gibbs free energy:

- $\Delta H = -49 \text{ (kJ mol}^{-1}\text{)}$; [1 mark]
- $\Delta S = -180 \text{ (J K}^{-1}\text{ mol}^{-1}\text{)}$; [1 mark]
- $\Delta G = -49 - (790 \times \frac{-180}{1000})$; [1 mark]
- $\Delta G = 93.2 \text{ (kJ mol}^{-1}\text{)}$; [1 mark]

[Total: 4 marks]

- To calculate ΔH :
 - $\Delta H^\ominus = \sum \Delta H_{\text{products}}^\ominus - \sum \Delta H_{\text{reactants}}^\ominus$
 - $\Delta H = (-201 + -242) - (-394)$
 - $\Delta H = -49 \text{ (kJ mol}^{-1}\text{)}$
- To calculate ΔS :
 - $\Delta S^\ominus = \sum S_{\text{products}}^\ominus - \sum S_{\text{reactants}}^\ominus$
 - $\Delta S = (238 + 189) - ((3 \times 131) + 214)$
 - Don't forget to multiply 131 by 3 as there are three moles of hydrogen present
 - $\Delta S = -180 \text{ (J K}^{-1}\text{ mol}^{-1}\text{)}$
- $-180 \text{ J K}^{-1}\text{ mol}^{-1}$ needs converting to kJ by dividing by 1000 so $-0.180 \text{ kJ K}^{-1}\text{ mol}^{-1}$
- The numbers are then substituted into the equation:
 - $\Delta G = \Delta H - T\Delta S$
- If you didn't convert ΔS to $\text{kJ K}^{-1}\text{ mol}^{-1}$, you could score the final mark for an answer of 142,151

1b

b) The reaction is:

- Not feasible as ΔG is positive / more than zero; [1 mark]

[Total: 1 mark]

- **Remember:** For a reaction to be feasible, ΔG should be negative

1c

c) To calculate the temperature at which the reaction becomes feasible:

- $T = \Delta H / \Delta S$; [1 mark]
- $T = -49 / -0.180$; [1 mark]
- The reaction becomes feasible at $T = 272 \text{ K}$

OR

$$T < 272 \text{ K}; [1 \text{ mark}]$$

[Total: 3 marks]

- As stated in part (b) for a reaction to be feasible:
 - ΔG should be negative:
 - $\Delta G = \Delta H - T\Delta S$
 - $0 < \Delta H - T\Delta S$
 - $\Delta H < T\Delta S$
 - $T < \Delta H / \Delta S$
 - $T < -49 / -0.180$
 - $T < 272 \text{ K}$
- This means that the **minimum** temperature that the reaction is feasible is 272 K
- You can use $=$ instead of $>$ to perform this calculation, i.e. assume $\Delta G = 0$ for the reaction to be feasible

1d

d) It has a negative entropy change because:

- Fewer molecules / moles of gas / product are formed (in the reaction); [1 mark]

[Total: 1 mark]

- For the reaction $\text{CO}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_3\text{OH}(\text{g}) + \text{H}_2\text{O}(\text{g})$
 - 4 molecules of gas produce 2 molecules of gas
 - The system becomes less disordered so entropy decreases
 - A decrease in entropy is shown by a negative value for entropy change

2a

a)

i) To show calcium carbonate is stable at 25 °C:

- $\Delta S = 40 + 214 - 89 = 165 \text{ (J K}^{-1} \text{ mol}^{-1})$; [1 mark]
- $\Delta G = +178 - (298 \times 0.165)$; [1 mark]
- $\Delta G = 129 \text{ kJ mol}^{-1}$; [1 mark]
- ΔG is above 0 / positive so the reaction is not feasible

ORAs ΔG is above 0 / positive, CaCO_3 is stable; [1 mark]

ii) To calculate the minimum temperature needed for calcium carbonate to decompose:

- $0 = \Delta H - T\Delta S$
OR
 $\Delta H = T\Delta S$
OR
 $T = \Delta H \div \Delta S$; [1 mark]
- $178 \div 0.165 = 1079 \text{ K}$
OR
 $806 \text{ }^\circ\text{C}$; [1 mark]

[Total: 6 marks]

- The first part of the question is basically asking you to show that the thermal decomposition of calcium carbonate is not feasible at 25 °C
- To determine this use, $\Delta G = \Delta H - T\Delta S$
- You are given ΔH in the question and need to calculate ΔS using:
 - $\Delta S^\ominus = \sum S_{\text{products}}^\ominus - \sum S_{\text{reactants}}^\ominus$
- Once calculated, your answer of $165 \text{ J K}^{-1} \text{ mol}^{-1}$ needs converting to $\text{kJ K}^{-1} \text{ mol}^{-1}$ by dividing by 1000
- You also need to convert 25 °C into K by adding 273
- **Remember:** If ΔG is above 0, i.e. positive, the reaction is not feasible
 - In this case, ΔG is 129 kJ mol^{-1} so the calcium carbonate will not decompose and is therefore stable
- Using your figures from part (i), you can calculate the temperature at which the

reaction becomes feasible by:

- Assuming $\Delta G = 0$
- Rearranging the equation to find T
- You **must** specify units in your answer to achieve the final mark

2b

b) The conditions need to be changed to the following:

- Increase the temperature; [1 mark]
- $T\Delta S > \Delta H$; [1 mark]
- $\Delta G < 0$; [1 mark]

[Total: 3 marks]

- $T\Delta S$ needs to be equal to or higher than ΔH for the reaction to become feasible

2c

c) This process is spontaneous because:

- The molecules become more disordered when evaporation occurs; [1 mark]
- The entropy change is positive; [1 mark]
- $T\Delta S > \Delta H$; [1 mark]
- (So) $\Delta G < 0$; [1 mark]

[Total: 4 marks]

- Entropy is a measure of disorder
- The higher the level of disorder the higher the entropy, so evaporation will have a positive entropy change as the arrangement of molecules becomes more disordered changing from a liquid to a gas
- ΔH is positive for an endothermic reaction
- For temperatures above a certain value, $T\Delta S$ will be greater than ΔH
- When $T\Delta S$ is greater than ΔH , ΔG will be less than 0
- Therefore, the reaction is feasible / spontaneous
- **Note:** If T falls below the value where $T\Delta S$ is greater than ΔH , i.e. $T\Delta S < \Delta H$, the reaction is no longer feasible

2d

d) To calculate the temperature above which the reaction is not feasible:

- $\Delta S = 189 - (205 \div 2 + 131) = -44.5 \text{ (J K}^{-1}\text{mol}^{-1})$; [1 mark]
- $T = \Delta H / \Delta S$; [1 mark]
- $\Delta H = -242 \text{ kJ mol}^{-1}$

AND

$$T = (-242 \times 1000) / -44.5; \text{ [1 mark]}$$

- 5438 K

OR

$$5165 \text{ }^{\circ}\text{C}; \text{ [1 mark]}$$

[Total: 4 marks]

- The equation for the reaction is:
 - $\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g})$
- **Remember:**
 - $\Delta S = \sum S_{\text{products}} - \sum S_{\text{reactants}}$
 - $\Delta H = \sum \Delta H_{\text{f}}(\text{reactants}) - \sum \Delta H_{\text{f}}(\text{products})$
- The enthalpy of formation for elements is 0 so in this case no calculation is required for ΔH
- When calculating ΔS , don't forget to divide 205 by 2 (this is a common mistake on exams)
- You could also have converted ΔS into $\text{kJ K}^{-1} \text{mol}^{-1}$ instead of ΔH into J mol^{-1}
- You must include units for the final marking point
- Values in the range of 5400 to 5500 K would be accepted

2e

e) If the sample was heated to higher than the temperature in part (d):

- (The water) would decompose into hydrogen and oxygen

OR

It would decompose into its elements; [1 mark]

- Because ΔG for this reaction would be less than 0

OR

The reverse reaction is feasible; [1 mark]

2e

e) If the sample was heated to higher than the temperature in part (d):

- (The water) would decompose into hydrogen and oxygen

OR

It would decompose into its elements; [1 mark]

- Because ΔG for this reaction would be less than 0

OR

The reverse reaction is feasible; [1 mark]

[Total: 2 marks]

- The formation of gaseous water from hydrogen and oxygen is a reversible reaction
- Above the temperature of 5438 K the forward reaction is no longer feasible, but the reverse reaction is because:
 - $T\Delta S$ will be greater than ΔH
 - ΔG will therefore be less than 0

3a

a) The completed table showing the sign of ΔS is:

chemical reaction or process	sign of ΔS
$\text{MgCl}_2(\text{s}) + (\text{aq}) \rightarrow \text{MgCl}_2(\text{aq})$	+
$\text{Ca}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{aq}) + \text{H}_2(\text{g})$	+
$\text{Na}_2\text{SO}_4(\text{s}) + 10\text{H}_2\text{O}(\text{l}) \rightarrow \text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}(\text{s})$	-
$2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g})$	-
$\text{C}_2\text{H}_5\text{OH}(\text{g}) \rightarrow \text{C}_2\text{H}_5\text{OH}(\text{l})$	-

- All 5 correct; [2 marks]
- 3 or 4 correct; [1 mark]
- 0, 1 or 2 correct; [0 marks]

[Total: 2 marks]

- **Remember:**
 - Entropy increases, i.e., ΔS is positive, as the system becomes less ordered /

more disordered

- Entropy decreases, i.e., ΔS is negative, as the system becomes more ordered / less disordered
- Use the state symbols to help understand how the order of the system is changing in each reaction / process
- $\text{MgCl}_2(\text{s}) + (\text{aq}) \rightarrow \text{MgCl}_2(\text{aq})$
 - MgCl_2 is dissolving in water, the magnesium and chloride ions spread through the water, there is an increase in the number of ways of arranging the energy and the order of the system decreases
 - Entropy increases
- $\text{Ca}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{aq}) + \text{H}_2(\text{g})$
 - Solid Ca reacts with an aqueous solution to form another aqueous solution and a gas
 - The order of the system decreases as the solid particles are fixed in positions in a regular arrangement whereas gas particles are able to move freely and are spread out
 - Entropy increases
- $\text{Na}_2\text{SO}_4(\text{s}) + 10\text{H}_2\text{O}(\text{l}) \rightarrow \text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}(\text{s})$
 - This shows the formation of a hydrated salt where a solid is formed from a solid and liquid
 - The order of the system increases as the liquid particles are able to move past each other whereas the product particles are fixed in positions in which they can only vibrate
 - Entropy decreases
- $2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g})$
 - 3 molecules of gas react to form 2 molecules of gas
 - Order of the system increases as there are fewer ways of arranging the energy of the system over the products
 - Entropy decreases
- $\text{C}_2\text{H}_5\text{OH}(\text{g}) \rightarrow \text{C}_2\text{H}_5\text{OH}(\text{l})$
 - This shows a change of state as a gas changes to a liquid
 - Order of the system increases as the molecules become closer together, the ability of the particles to move decreases and there are fewer ways to arrange the energy
 - Entropy decreases

3b

b) The feasibility of this reaction will change as the temperature decreases as follows:

- As temperature decreases, ΔG decreases

AND

The reaction is more feasible; [1 mark]

- Because $T\Delta S$ decreases in magnitude / becomes less negative; [1 mark]

[Total: 2 marks]

- This question relies on your knowledge of the equation $\Delta G = \Delta H - T\Delta S$
- The value of ΔG , and therefore whether the reaction is feasible, depends on the value of ΔH and ΔS , and T :

If ΔH	And if ΔS	Then ΔG	Feasible?
Is negative	Is positive	Always negative	Always
Is positive	Is negative	Always positive	Never
Is negative	Is negative	Negative at low temperatures	Feasible at low temperatures
Is positive	Is positive	Negative at high temperatures	Feasible at high temperatures

- So, in this case, as the value of ΔH is negative and the value for ΔS is negative, the value for ΔG will be negative but only when T is low
- Students often struggle with this concept and it can be useful to think of this numerically so you can see what happens to the value of ΔG when T decreases
- To do this, you need to choose imaginary values for ΔH and ΔS and two different temperatures - the actual values don't matter so choose numbers that are really easy to perform calculations with
 - e.g. $\Delta H = -100$, $\Delta S = -10$, $T_1 = 100$, $T_2 = 200$
- Then calculate ΔG for both temperatures
 - ΔG at $T_1 = -100 - (100 \times -10) = 900$
 - ΔG at $T_2 = -100 - (200 \times -10) = 1900$
- This shows that when the temperature decreases, ΔG decreases

3c

c) To determine the temperature at which reaction **A** becomes feasible:

- $\Delta H_f = \sum \Delta H_f \text{ products} - \sum \Delta H_f \text{ reactants}$; [1 mark]

- $\Delta H_f = (2 \times -395.7) - (2 \times -296.8 + 0) = -197.8 \text{ (kJ mol}^{-1}\text{)}; [1 \text{ mark}]$
- $\Delta S = \Sigma \Delta S_{\text{products}} - \Sigma \Delta S_{\text{reactants}}; [1 \text{ mark}]$
- $\Delta S = (2 \times 256.8) - (2 \times 248.2 + 205.1) = -187.9 \text{ (J mol}^{-1}\text{)}; [1 \text{ mark}]$
- Convert ΔS from $\text{J K}^{-1} \text{ mol}^{-1}$ to $\text{kJ K}^{-1} \text{ mol}^{-1}$ / $\Delta S = 0.1879 \text{ (kJ K}^{-1} \text{ mol}^{-1}\text{)}; [1 \text{ mark}]$
- $T = \frac{\Delta H}{\Delta S} = \frac{197.8 \text{ kJ mol}^{-1}}{0.1879 \text{ kJ K}^{-1} \text{ mol}^{-1}} = 1053 \text{ (K)}; [1 \text{ mark}]$

[Total: 6 marks]

- You should know all of the equations to calculate ΔH , ΔS and ΔG
- **Careful:**
 - The value for the formation of oxygen is zero when calculating enthalpy change as it is still an element in its standard state in the reaction
 - The value for the formation of oxygen is not zero when calculating entropy
- In order to calculate the temperature at which the reaction becomes feasible, ΔG should be negative
- A reaction is not feasible if ΔG is greater than 0
- Therefore by rearranging the equation for the case where $\Delta G = 0$ the temperature at which the reaction becomes feasible can be found
- Do not forget to make sure both ΔH and ΔS are given in J or kJ, **either**
 - Multiply $\Delta H^\ominus$ by 1000
 - OR**
 - Divide $\Delta S^\ominus$ by 1000 (more commonly done)
 - This will make sure they are in the same unit

